

Note: There are FIVE questions, answer ALL of them.

You have necessary CSV/caption files that each contain cleaned comments and captions collected earlier in this projec. Your job is to explore these datasets carefully and produce structured outputs that highlight both important words and patterns of words. Follow the tasks below:

Q. No.	Question Description
1.	Build the analysis-ready comment corpus Load your cleaned comments dataset into a pandas DataFrame. Check that the cleaned token column is usable for analysis, and convert each row into a single text string that can be used for machine learning. Remove rows with missing or empty text after conversion. Then print: • the number of usable comments • the name of the text field you used • one sample processed comment text
2.	Vectorize the comments and choose a suitable number of clusters Convert the processed comments into TF-IDF features using a reasonable feature limit such as 1500 . Then apply the Elbow Method for k = 2 to 6 and plot the inertia values. Based on the plot, choose one value of k and justify your choice in 1 to 2 sentences.
3.	Group the comments and interpret the cluster themes [2] Run K-Means using your chosen value of k . For each cluster: • report the number of comments in the cluster • print the top 5 representative terms Then write a short interpretation for any two clusters , explaining what kind of audience discussion each cluster seems to represent.

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4.	Compare sentiment across the discovered groups Use a sentiment analysis tool such as TextBlob or VADER on the comments. For each cluster, calculate how many comments are positive, neutral, and negative. Present the result as either: • a table, or • a bar chart Then identify which cluster appears most positive or most negative, and state one possible reason.
5.	Visualize the cluster structure and write a final observation Reduce the TF-IDF feature space to 2 dimensions using PCA and create a scatter plot of the comments colored by cluster label. After plotting, write a short final observation of 3 to 4 sentences covering: • whether the clusters look clearly separated or partially mixed • whether the visual pattern supports your chosen value of k • one practical limitation of clustering short YouTube comments